

Ahmed Fathy

Software Engineer

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TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, JavaScript, TypeScript, SQL, PHP, Java, Assembly (x86/MIPS), OpenGL, GLSL
- **Frontend & Full-Stack Web:** React.js, Next.js, Redux Toolkit, Tailwind CSS, Three.js, Node.js, Express.js, RESTful APIs, GraphQL, PostgreSQL, MongoDB, Laravel
- **AI & Machine Learning:** PyTorch, TensorFlow, Keras, OpenCV, YOLO, Deep Learning, CNNs, Image Processing, NumPy, Pandas, Scikit-learn
- **Robotics, IoT & Systems:** ESP32, Arduino, Unix/Linux Development, Multithreading, Memory Management, Network Socket Programming, WebSockets
- **DevOps & Core Concepts:** Docker, Git Version Control, CI/CD Pipelines, Object-Oriented Programming, Microservices, System Architecture, Algorithms

EXPERIENCE

- **Al-Nada Scientific Office** | *Software Developer (Part-time)* | *Cairo, Egypt* | *June 2022 – 2025*
 - **ERP System:** Developed a local desktop application (packaged with Electron) managing customers, suppliers, inventory, and sales records using React.js, Express.js, and PostgreSQL.
 - **Corporate Web Solution:** Built the official company website integrated with an e-commerce platform designed to seamlessly manage and showcase sponsors, partners, services, and dynamic product catalogs.

EDUCATION

- **Cairo University, Faculty of Engineering** | *Giza, Egypt*
B.Sc. in Computer Engineering; GPA: 3.7 | *2022*

PROJECTS

- **TCCD Official Website & Management System** | *Frontend Development: React, TypeScript* | *Live Preview* | *2026*
 - **Impact:** Developed a comprehensive student activity platform that successfully scaled to support and manage over 20,000 active users.
 - **Core Features:** Engineered a robust event and judging system that handles registration slots, waiting lists, and distinct role-based access for members, standard users, and corporate entities.
 - **Administration:** Built an advanced admin dashboard equipped with user-facing analytics to streamline the management of events, user data, and company profiles.
- **Image Restoration Pipeline** | *Deep Learning & Computer Vision: Python* | *Source Code* | *2024*
 - **Pipeline:** Constructed end-to-end restoration pipeline: Denoising, Deblurring, and Inpainting.
 - **Algorithms:** Applied Telea (FMM) inpainting and deep learning models to restore damaged historical photographs.
- **Tactic (Robotics & AI)** | *Embedded Systems* | *Source Code* | *2024*
 - **System Integration:** Developed browser-controlled robotics suite integrating AI computer vision with low-latency hardware control.
 - **Firmware:** Programmed ESP32 microcontrollers to handle WebSocket communication for real-time motor control and telemetry.
 - **Backend Processing:** Deployed YOLOv5 and OpenCV on host server to process visual feedback and automate robotic decision-making.

STUDENT ACTIVITIES

- **IEEE | UI/UX Design & Frontend Team Leader** | *Cairo* | *Oct 2023 – Present*
 - **Portfolio Platform Development:** Engineered comprehensive web platform featuring workshop and event management systems with admin dashboard for content creation, role-based access control enabling instructors to upload course materials and students to register for workshops and events.
- **Technical Center for Career Development (TCCD)** | *Frontend Developer* | *Cairo* | *June 2025 – Present*
 - **Portfolio Platform Development:** Built an integrated portfolio, event management, HR, and judging system with role-based access, dashboards, analytics, attendance tracking, custom forms, and evaluation workflows to streamline operations and decision-making.
- **NASA Space Apps Challenge** | *Participant Project: "Orrey"* | *Cairo* | *Oct 2024*
 - **Orrey App:** Developed an interactive Orrey app visualizing near-Earth objects using React and Three.js.